

IEPS: Programming Assignment 3

Explainable Retrieval-Augmented Generation

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I. INTRODUCTION

This assignment extends our PA2 Slovenian information-retrieval system into an explainable Retrieval-Augmented Generation (RAG) pipeline. We retain the successfully evaluated MedOverNet corpus and retrieval configuration, and add answer generation in two directly comparable modes: *With Context*, where retrieved chunks form explicit evidence, and *Without Context*, where the same local language model receives only the question.

The implementation is intentionally experimental rather than deployed as a user interface. For every RAG answer it stores the evidence passages, original vector ranks, reranked ranks and scores, allowing answer errors to be related to retrieval failures rather than treated as opaque model behavior.

II. IMPLEMENTATION SUMMARY

A. Existing PA2 Evidence Store

PA3 reuses the PA2 PostgreSQL/pgvector database without schema changes or re-indexing. The corpus contains Slovenian health and wellness material from MedOverNet, including editorial articles and forum threads. PA2 created `page_segment_long` chunks: articles are grouped into approximately 250 words with 50-word overlap; forum content is kept close to individual posts using approximately 220 words with 30-word overlap.

This was the appropriate continuation of PA2: long segments preserve enough semantic context for a generated response, while PA2 manual evaluation showed that the 50-character comparison chunks frequently lose meaning at chunk boundaries.

B. Models and Libraries

Queries are embedded using `sentence-transformers/LaBSE`, a multilingual sentence embedding model whose 768-dimensional output exactly matches the already populated `vector(768)` columns. Candidate segments are obtained using cosine distance through pgvector and then reranked with `BAAI/bge-reranker-v2-m3`, the multilingual cross-encoder already evaluated in PA2.

Answers are generated through a local Ollama service using `gemma3:4b`. The model has approximately four billion parameters; its Ollama package is approximately 3.3 GB and its documented multilingual support makes it a practical local model for Slovenian questions [1]. A deterministic experimental setting is used with temperature set to zero.

The implementation is divided into:

- `retriever.py`: LaBSE query embedding, cosine retrieval of 20 candidate chunks, and reranking to the final top five evidence chunks;
- `prompts.py` and `generator.py`: separate prompts for evidence-grounded and LLM-only answering, and the Ollama API integration;
- `rag_pipeline.py`: single-question command-line interface;
- `evaluate.py`: batch comparison runner storing full JSON traces and manual scoring fields.

C. Prompt Design

In RAG mode, the prompt includes numbered chunks and source URLs. The model is instructed to use only that evidence, cite used chunks as [1], [2], and explicitly state when the context is insufficient. This is particularly important for medical content and out-of-domain questions.

In LLM-only mode, no chunks or URLs are provided. The model is asked for a short Slovenian response and told not to invent references. Keeping the question and generation model fixed while changing only the evidence enables a transparent comparison.

III. SYSTEM EVALUATION CRITERIA

A. Data Processing

No new data preparation is performed in PA3. The evaluation relies on the PA2 decision to use logical long chunks with overlap. Data-processing quality is therefore assessed through inspectability of the retrieved chunk: a good chunk should contain an interpretable passage that directly supports an answer rather than only a matching keyword.

B. Retrieval

For each query, LaBSE cosine search fetches 20 initial candidates from `page_segment_long`. The cross-encoder reranker chooses five passages supplied to the LLM. We store the vector rank, cosine distance, final rank, reranker score, text and URL for every final passage.

Retrieval relevance is scored manually on a three-point scale: 0 (not informative), 1 (partially informative), and 2 (directly supports answering the question). For each answer we also assess answer quality, groundedness in the visible passages, and whether the evidence makes the result diagnosable.

C. Failure Categories

Failures are separated into four types: irrelevant retrieval, insufficient corpus coverage, ambiguous query, and generation hallucination. This distinction prevents attributing an incorrect answer to the generator when the index never contained suitable evidence.

IV. EVALUATION QUERY SET

Table I lists the nine controlled evaluation questions. The first three good queries preserve PA2 comparability. The additional topics were selected only after they were observed in stored PA2 extracted content or PA2 retrieval outputs. The three adverse cases are retained from PA2 because they expose ambiguity and corpus-boundary failures.

Table I
PA3 CONTROLLED QUERY SET AND INTENDED DIAGNOSTIC ROLE

Type	Query	Coverage basis / expected behavior
Good	Kateri so simptomi visokega krvnega tlaka in zakaj je merjenje pomembno?	PA2 reranking surfaced a direct hypertension-management article.
Good	Kako zdrav življenjski slog pomaga pri sladkorni bolezni tipa 2?	PA2 surfaced a diabetes and cardiometabolic lifestyle article.
Good	Kako izboljšati spanec in zakaj je reden spanec pomemben?	PA2 retrieval and validation content contain sleep-focused articles.
Good	Kateri so značilni simptomi Parkinsonove bolezni in kako pomaga gibanje?	The extracted PA2 validation sample contains a Parkinson disease article.
Good	Kateri znaki lahko kažejo na pomanjkanje vitamina B12?	The extracted PA2 validation sample contains a vitamin B12 article.
Good	Zakaj so beljakovine pomembne pri ohranjanju mišične mase?	A saved PA2 retrieval run contains protein and muscle-preservation passages.
Bad	Boli me	Underspecified symptom; retrieval cannot establish the user’s intended pain.
Bad	Kakšne so pravne posledice neplačila prometne kazni v Sloveniji?	Legal question outside the medical corpus; RAG should decline evidence-based answering.
Bad	ATM	Unspecified acronym; PA2 already showed unrelated retrievals and very low reranker scores.

V. EVALUATION RESULTS

The final run was executed on May 30, 2026, with the fixed retrieval and generation configuration described above. The restored PA2 evaluation database contained 36,816 long segments with embeddings. The complete trace, including all retrieved chunks and full generated answers, is stored in `pa3/rag/runs/20260530T022922Z_evaluation.json`.

Table II summarizes the generated answers and manual scores. The score order is retrieval relevance / RAG answer quality / LLM-only answer quality. In the six good cases, retrieval relevance reached the maximum score of 12/12. RAG answer quality reached 10/12, while LLM-only answers reached 6/12. Across all nine cases, RAG answer quality was 15/18 and LLM-only answer quality was 7/18.

VI. DISCUSSION AND LIMITATIONS

The results show the intended trade-off clearly. When the query matches the medical corpus, RAG improves answer quality by adding concrete Slovenian evidence and making each claim inspectable. The blood-pressure, sleep, vitamin B12 and protein queries are the strongest examples: the LLM-only answers are fluent enough to sound plausible, but they are shorter, less specific, and sometimes linguistically unreliable.

Retrieval fails in two different ways. First, a query can be underspecified: “Boli me” and “ATM” do not provide enough intent, so reranking cannot recover a stable answer from the corpus. Second, the corpus can be out of domain: the traffic-fine query retrieves weak legal/forum fragments from a medical site, but not enough evidence for a legal answer. In those cases, the best RAG behavior is not a better answer, but a transparent refusal.

Generation errors still occur after good retrieval. In the diabetes and Parkinson cases, relevant chunks were retrieved, but the model introduced awkward or weakly grounded wording. This confirms that retrieval quality constrains the answer, but does not fully guarantee generation quality.

The primary experimental limitation is qualitative assessment: nine questions provide interpretable case analysis rather

than a statistically representative benchmark. This is appropriate to the assignment’s focus on transparency, but it should not be interpreted as clinical performance evaluation.

VII. CONCLUSION

PA3 extends the strong PA2 retrieval system without disrupting its documented choices. It adds a local Ollama generation layer, an evidence-restricted RAG prompt, an LLM-only comparison mode, and complete retrieval traces for evaluation. The central explainability benefit is concrete: every RAG answer can be checked against the passages that shaped it, and failures can be classified as missing evidence, poor retrieval, ambiguous input, or model generation error.

REFERENCES

- [1] Ollama, “gemma3 model library,” <https://ollama.com/library/gemma3>, accessed May 25, 2026.

Table II
COMPARISON OF RAG AND LLM-ONLY ANSWERS IN THE FINAL PA3 EVALUATION

Type	Query	RAG answer summary	LLM-only answer summary	Score	Comment
Good	Blood pressure and symptoms measurement	Correctly explains that hypertension is often asymptomatic, lists severe warning symptoms, and ties measurement to prevention of stroke, heart and kidney complications.	Mentions asymptomatic pressure and complications, but contains incorrect or very poor wording.	2/2/1	Retrieval adds direct evidence for why measurement matters.
Good	Type 2 diabetes and lifestyle	Uses relevant evidence about diet, physical activity, weight loss, glucose control and possible remission.	Gives a short generic answer about diet, activity and insulin sensitivity.	2/1/1	Retrieval succeeds; one generated sentence about regular treatment is poorly grounded.
Good	Sleep quality and regular sleep	Grounds the answer in consistent sleep schedule, circadian rhythm, sleep quality and health risks of chronic sleep deprivation.	Lists general sleep-hygiene advice with weak Slovenian phrasing.	2/2/1	RAG adds concrete reasons and visible sources.
Good	Parkinson symptoms and exercise	Finds the Parkinson article and explains movement, balance, boxing and social exercise support.	Provides only a brief generic description of symptoms and movement therapy.	2/1/1	Evidence is relevant, but generated terminology is sometimes awkward.
Good	Vitamin B12 deficiency signs	Uses B12-specific passages about tongue changes, mouth symptoms, paresthesias and neurological issues.	Lists common signs, but remains generic and has language errors.	2/2/1	RAG improves precision and source traceability.
Good	Protein and muscle mass	Explains amino acids, muscle preservation during weight loss and the need to combine protein intake with strength training.	States only that proteins are building blocks for muscle growth and repair.	2/2/1	Retrieval adds the key nuance that protein alone is not enough.
Bad	Boli me	Shows several unrelated pain forum cases and cautiously states that diagnosis is not possible from the evidence.	Responds empathetically and advises medical help, but gives no explanation.	1/1/1	Query is too ambiguous for reliable retrieval.
Bad	Legal consequences of unpaid traffic fine	Correctly refuses to infer a legal answer because retrieved sources do not cover the question.	Gives confident legal claims without evidence.	0/2/0	Medical corpus has insufficient legal coverage; refusal is correct.
Bad	ATM	Shows unrelated financial/spam forum chunks and refuses to answer the underspecified acronym.	Hallucinates an incorrect Slovenian expansion and description.	0/2/0	Ambiguous acronym produces irrelevant retrieval; RAG remains diagnosable.